Jingchun Shao

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Education

Texas A&M University, Ph.D. in Mathematics — Aug 2025 – Jun 2029 (expected)

- **GPA:** 4.00/4.00; passed qualifying exams in analysis and applied mathematics.
- **Relevant coursework:** Statistical Learning Theory and Empirical Processes; Compressive Sensing; Numerical PDEs.
- **Advisor:** Simon Foucart
- **Fellowship:** Joel Zinn Memorial Endowed Fellowship (2025)

University of Chicago, M.S. in Computational and Applied Mathematics — Sep 2023 – Jun 2025

- **GPA:** 3.83/4.0; **M.S. thesis:** *Quantized Tensor Train for Compressed Sensing* (advisor: Yuehaw Khoo).
- **Relevant coursework:** Machine Learning on Manifolds; Solving PDEs with Machine Learning; Convex Optimization; Matrix Computation; Fast Algorithms.

Hunan University, B.S. in Pure and Applied Mathematics — Sep 2019 – Jun 2023

- **Honors:** School-Level Scholarship; Mitacs Scholarship (exchange at the University of Alberta)

Research Experience

Research Assistant, Texas A&M University — Advisor: Simon Foucart — Aug 2025 – Present

- Research on scalable tensor methods and optimal recovery for high-dimensional scientific computing.

Research Assistant, University of Chicago — Advisor: Yuehaw Khoo — Sep 2023 – May 2025

- **Operator learning:** Trained a score-based generative model using a GPU to learn the Ginzburg–Landau Markov operator; achieved $\sim$5% relative error on a second-moment estimate in 64 dimensions.

Research Intern, Chinese Academy of Sciences — Supervisor: Shihua Zhang — Jul 2021 – Jun 2022

- Studied machine-learning approaches to discovering and solving PDEs, with emphasis on physics-informed neural networks (PINNs).

Manuscripts

Compressed Sensing with Quantized Tensor Trains (QTTs) [arXiv] [Code]
Jingchun Shao. *Submitted, 2026.*

- Proves recovery guarantees for high-dimensional signals with low-rank tensor-train representations; develops a structure-aware initializer that improves iterative reconstruction.
- **Best Poster Award** — Fall Fourier Talks, University of Maryland, Oct 2026.

Worst-Case Completion of Tensors with Approximately Few ANOVA Terms [arXiv] [Code]
Simon Foucart and Jingchun Shao. *Submitted, 2026.*

- Studies worst-case tensor completion when low-order ANOVA interactions dominate; develops FFT-based projections and efficient numerical routines to make optimal recovery scalable.

Selected Experience and Awards

Big Quantum Hackathon, Chicago — Sep 2023

- **1st place (business phase); 3rd place (technical phase).** Modeled Boeing wing-ply stacking as an optimization problem and used CUDA Quantum to test quantum relaxations of the model.

Quantitative Research Intern, Yanhe Fund – Shanghai, China — Jun 2022 – Aug 2022

- Used Python and NumPy to validate a convertible-bond strategy later deployed as a fund product.

Technical Skills

Python (NumPy, SciPy, PyTorch), C++, MATLAB, Java, LaTeX